Wackiest USB drives in the markets

warranty, and is willing to spend a little more to get some peace of mind.

I divided the flash drives into three categories: 8, 16 and 32 GB. Unfortunately, one of the drives, the 16 GB Corsair Flash Voyager GT, was obviously faulty. One would assume it would outperform its cousin, the Corsair Flash Voyager, by a mile. When testing, however, this drive gave us the lowest scores of all the drives in the test, and we just had to leave it out. We'll try and make sure to get another GT and include it in our Bazaar section in the near future.

Anyway, back to the test. Although, like a good little reviewer, I did use HD Tach to test all storage components we received, the scores are often very deceptive for flash memory. HD Tach is specially coded to bypass operating system overheads, and checks for read and write speeds from different areas of the drives. This is generally needed for drives that have spinning parts, where read and write speeds vary, based on how far the sensor is from the centre of the platter. HD Tach also works best on non-formatted drives on Windows XP, but we decided that our test rig should run Vista because of the high rate of adoption (forced, or otherwise) that Vista now has; and we couldn't delete all formatting information on the flash memory. For those of you interested in soft scores, we've retained them for the Flash drives, but not included them in the decision making process.

8 GB

Plump portables

This is probably the most popular category right now, because of the drastic fall in prices, and the numbers show. We had a drive each from Apacer, Corsair and SanDisk, and two from Kingston. Before I get into the reviews, I should point you to the tables, because that's where you will find all the numbers. The reviews below will give you the information that the numbers cannot.

SanDisk Ultra Cruzer Titanium

The 8 GB segment seems to have the most variety, in terms of looks, with quite a few eye-catchers here. By far the one that stands out the most is the SanDisk Cruzer Titanium. This slick looking, shiny silver drive actually has a titanium casing that SanDisk claims is "crush-resistant". Obviously I haven't tested this particular quality (and continue to fight my desire to run it over with a car), but it does feel really sturdy.

In terms of performance, this drive had the best assorted read speed, clocking in at an average of 24.4 Mbps – transferring 4 GB of assorted data in just 168 seconds. Now this is impressive, because all drives seem to slow down considerably in our assorted file transfers – all except this one, though. However it did get beaten in the three other tests, by Kingston's HyperX.

SanDisk Ultra Cruzer Titanium

The only grouse I have with the build quality is the plastic slider mechanism, which although it feels as sturdy as any of the other sliders, just does not sit well with the rest of the body construction. A nice metal slider would just make this the perfectly built drive. That said, the plastic slider's quality is top-notch. My personal preference is towards non-slider flash drives, but the Ultra Cruzer has forced me to rethink that.

In terms of software support, this drive pretty much has everything you need – the U3 platform assures you of this. If you're the clumsy type, this is a must buy, because it feels like nothing but a frustrated fling at a concrete wall could scratch this beauty. It's price of Rs. 1,449 is surprising, because it looks more expensive. When you consider the performance you get for such a low price, it's the obvious pick for our Best Buy award.

Corsair Flash Voyager

I'll be honest, I was actually looking forward to reviewing this drive's bigger brother – the Voyager GT. That drive was fried, sadly, so I popped open this one's packaging. This drive is distinctive, and looks good. The black and green look suits it. As soon as I picked it up, I could tell that this drive is built solidly, with a soft rubber sheath. The rubber quality seems pretty good, and, apologies to the Corsair guys, but I just had to drop test this one. To answer the questions that some of you might have: yes, it does bounce; a little oddly, like a crazy ball; and is a real joy to play hacky-sack with. That should tell you all you need to know about its build quality. Funny how the Voyager kept functioning normally after all my abuse, and the Voyager GT didn't work out of the box!

Corsair Flash Voyager

HOW WE TESTED

Test Rig
For both tests, we used a PC running Windows Vista Ultimate, with only the essential services running, with the following configuration:

Intel Core2Duo E6700
2 GB AData DDR2 RAM
Seagate 7200.11 500GB
Gigabyte GA-965P-DQ6

Flash drives
All flash drives were connected to the test PC on the same USB port. Because you can't delete a flash drive's partition data like you can on a hard drive, you can't run HD Tach's write test, but this is really no big loss, because HD Tach proved to be a better indicator of internal hard drive speeds, and not so much for USB drives. Also, once the tests were run, it became apparent that HD Tach rarely mirrors real world results. If you look at the comparison table, you will notice that the HD Tach scores are pretty much identical across the board, with all drives offering similar results. However, the real world tests give very varied results. We've preserved this information for the curious amongst you, but have not really given any importance to anything but real world tests and build quality.

Since these are all large capacity flash drives, we formatted all of them to the NTFS file system, and used 4 GB of assorted and sequential data to test them. This was done because, when you have between 8 and 32 GB of storage space, you're going to use this for large movies or videos, and will often find yourself transferring GBs of data to and from the drive.

Memory Cards
This one's even simpler to call. All that matters here is real-world performance, and price. There are no features, build quality or anything else to speak of. We used a Kingston Media Reader to conduct the tests on all of the cards. Since the memory cards ranged from 1 GB to 32 GB, we used test files that were 512 MB in size. There were 2 basic tests that we ran – assorted and sequential – and we noted down scores for read and write for each test. We were careful to restart the test rig each time we finished one test, because we didn't want the small test files to be cached in memory.

You will experience this when transferring files to your memory card, and then copying them back to the PC – you will get transfer speeds of over 190 Mbps, because it's writing directly from memory to the PC hard drive. So don't get fooled into believing that your memory card is capable of such speeds if this happens to you.

We ran each test multiple times, and only if the results differed by less than five per cent, they were logged. This is done to ensure that operating system overheads are nullified.

Interpreting the scores is not that simple however, because of the usage scenarios of these cards. A detailed explanation is provided in a box next to each category.